
CURRENT AND PENDING (OTHER) SUPPORT INFORMATION

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person.

*NAME: Skjellum, Anthony null

PERSISTENT IDENTIFIER (PID): <https://orcid.org/0000-0001-5252-6600>

*POSITION TITLE: Professor

*ORGANIZATION AND LOCATION: Tennessee Technological University, Cookeville, Tennessee,
United States

Proposals and Active Projects

***Proposal/Active Project Title:** Collaborative Research: Conference: NAIRR Pilot
Expansion: FA1: Mid-South Conferences on
CyberInfrastructure Advances to Enable
Interdisciplinary AI Research in Health Disparities

***Status of Support:** Current

Proposal/Award Number:

***Source of Support:** National Science Foundation

***Primary Place of Performance:** Cookeville, TN (TN TECH)

***Proposal/Active Project Start Date: (MM/YYYY):** 10/2025

***Proposal/Active Project End Date: (MM/YYYY):** 04/2027

***Total Anticipated Proposal/Project Amount:** \$125,133

* **Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2025	0.25
2026	0.13

***Overall Objectives:**

The goal of the proposed conference series is to provide training, guidance, and support to the formation and furtherance of interdisciplinary research at the nexus of Advanced CyberInfrastructure, health disparities, and the domain sciences underlying the new and innovative solutions, including AI (deep learning, LLMs, etc.), which is a timely and vital area that needs to be advanced for the good of a large portion of the population in the Southeastern US and beyond.

***Statement of Potential Overlap:**

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No overlap.

***Proposal/Active Project Title:** Collaborative Research: CSSI: Frameworks: MPI4AI: Enhancing Performance and Productivity of AI Science through Next-generation High Performance Communication Abstractions

***Status of Support:** Current

Proposal/Award Number:

***Source of Support:** National Science Foundation

***Primary Place of Performance:** TN TECH, Cookeville, TN

***Proposal/Active Project Start Date: (MM/YYYY):** 09/2025

***Proposal/Active Project End Date: (MM/YYYY):** 09/2029

***Total Anticipated Proposal/Project Amount: \$1,200,000**

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2025	1
2026	1
2027	1
2028	1

***Overall Objectives:**

Research and development of improvements and extensions to the MPI standard and the Open MPI implementation to better support workloads related to machine learning and artificial intelligence.

***Statement of Potential Overlap:**

No overlap.

***Proposal/Active Project Title:** PSAAP-IV (FIC): Center for Optimized Modern Parallel Adaptive System Software (COMPASS)

***Status of Support:** Current

Proposal/Award Number:

***Source of Support:** University of New Mexico (DOE Subcontractor)

***Primary Place of Performance:** TN TECH, Cookeville, TN

***Proposal/Active Project Start Date: (MM/YYYY):** 08/2025

***Proposal/Active Project End Date: (MM/YYYY):** 09/2030

***Total Anticipated Proposal/Project Amount:** \$1,250,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.31
2027	0.31
2028	0.31
2029	0.72
2030	0.6

***Overall Objectives:**

The overarching objective of COMPASS is to increase the performance of parallel applications and exascale systems of interest and relevance to DOE/NNSA by enabling applications and system software to effectively optimize, tune, and adapt the range of software and hardware resources they use.

***Statement of Potential Overlap:**

No overlap.

***Proposal/Active Project Title:** Collaborative Research: CIRC: New: Next Generation Message-Passing Parallel Programming for Heterogeneous Architectures

***Status of Support:** Current

Proposal/Award Number: 210808

***Source of Support:** National Science Foundation

***Primary Place of Performance:** Cookeville, TN

***Proposal/Active Project Start Date: (MM/YYYY):** 06/2025

***Proposal/Active Project End Date: (MM/YYYY):** 05/2028

***Total Anticipated Proposal/Project Amount:** \$826,099

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2025	1
2026	1
2027	1

***Overall Objectives:**

The key objective of this grant is to extend the Message Passing Interface with practical cyberinfrastructure derived in part from our previous research funded by NSF and DOE. MPI Advance will provide researchers with the ability to support faster and more effective performance-portable parallel programming by leveraging their baseline MPI plus the new and modified algorithms, APIs, and procedures offered by this software. Significant efforts in training, outreach, and engagement are planned.

***Statement of Potential Overlap:**

No overlap.

***Proposal/Active Project Title:** CC* Strategy-Region: TREN: Building Tennessee Research and Education Network

***Status of Support:** Current

Proposal/Award Number:

***Source of Support:** National Science Foundation

***Primary Place of Performance:** TN TECH, Cookeville, TN

***Proposal/Active Project Start Date: (MM/YYYY):** 10/2024

***Proposal/Active Project End Date: (MM/YYYY):** 09/2026

***Total Anticipated Proposal/Project Amount:** \$200,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2025	0.25
2026	0.25

***Overall Objectives:**

Establish collaborations and discussions between entities in the state of Tennessee in pursuit of a regional high-performance research network.

***Statement of Potential Overlap:**

No overlap.

***Proposal/Active Project Title:** Collaborative Research: Beginnings: Creating and Sustaining a Diverse Community of Expertise in Quantum Information Science (EQUIS) Across the South-Eastern United States

***Status of Support:** Current

Proposal/Award Number:

***Source of Support:** National Science Foundation

***Primary Place of Performance:** TNTECH, Cookeville, TN

***Proposal/Active Project Start Date: (MM/YYYY):** 07/2023

***Proposal/Active Project End Date: (MM/YYYY): 06/2026**

***Total Anticipated Proposal/Project Amount: \$258,486**

* Person Months per budget period Devoted to the Proposal/Active Project:

Year	Person Months
2023	0.01
2024	0.01
2025	0.01

***Overall Objectives:**

Workforce development in Quantum computing in a 5-university consortium under the ExLent RFP.

***Statement of Potential Overlap:**

No overlap.

***Proposal/Active Project Title:** CICI: IPAAI: Provenance-Aware AI Workflows: Building Trust and FAIR-ness in AI Datasets and Models

***Status of Support:** Pending

Proposal/Award Number:

***Source of Support:** National Science Foundation

***Primary Place of Performance:** Tennessee Technological University, Cookeville, TN

***Proposal/Active Project Start Date: (MM/YYYY):** 10/2026

***Proposal/Active Project End Date: (MM/YYYY):** 09/2029

***Total Anticipated Proposal/Project Amount:** \$900,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2027	0.55
2028	0.55
2029	0.55

***Overall Objectives:**

This project aims to develop an opt-in, blockchain- and smart-contract-based cyberinfrastructure that enhances the integrity and provenance of AI datasets and models. The system will support provenance-aware registration, verification, and large-file transfer within Python-based AI workflows, while integrating with major repositories such as Hugging Face, Kaggle, and Mendeley Data. By enabling end-to-end provenance tracking, model-dataset linkage, community feedback, and post-hoc impact analysis, the project aims to enhance trust in AI-enabled scientific results and promote provenance-enhanced workflows as a community best practice.

***Statement of Potential Overlap:**

This proposal.

***Proposal/Active Project Title:** AI-Driven Communication Optimization and Synthesis for Mission HPC Applications

***Status of Support:** Pending

Proposal/Award Number:

***Source of Support:** DOE

***Primary Place of Performance:** Tennessee Tech, Cookeville, TN, USA

***Proposal/Active Project Start Date: (MM/YYYY):** 07/2026

***Proposal/Active Project End Date: (MM/YYYY):** 03/2027

***Total Anticipated Proposal/Project Amount:** \$750,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
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Year	Person Months
2026	1
2027	0.5

***Overall Objectives:**

This project addresses DOE Genesis Mission Focus Area 18-A (AI-Driven Code Porting and Optimization) by developing and demonstrating an AI-driven iterative pipeline to optimize communication in production HPC code.

***Statement of Potential Overlap:**

No overlap.

***Proposal/Active Project Title:** AI-Driven Observable Synthesis for the INQ Electronic Structure Engine via Large Language Models: Structure Engine via Large Language Models: Accelerated Materials Discovery for the Genesis Mission Accelerated Materials Discovery for the Genesis Mission

***Status of Support:** Pending

Proposal/Award Number:

***Source of Support:** DOE

***Primary Place of Performance:** Tennessee Tech, Cookeville, TN

***Proposal/Active Project Start Date: (MM/YYYY):** 07/2026

***Proposal/Active Project End Date: (MM/YYYY):** 03/2027

***Total Anticipated Proposal/Project Amount:** \$750,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	1
2027	0.5

***Overall Objectives:**

This Phase I Genesis project responds to Topic 18-B with a transformative approach to enhancing DOE-relevant applications. Investigators at TN TECH, LLNL, and UT Chattanooga will explore revolutionary approaches to using AI to generate new observables in an important LLNL Quantum Materials Code (INQ). The results of this work will be generalizable across the DOE complex of large-scale applications.

***Statement of Potential Overlap:**

This proposal.

***Proposal/Active Project Title:** AI-Driven Runtime Consciousness for Exascale Performance to Accelerate Scientific Discovery

***Status of Support:** Pending

Proposal/Award Number:

***Source of Support:** DOE

***Primary Place of Performance:** Tennessee Tech, Cookeville, TN

***Proposal/Active Project Start Date: (MM/YYYY):** 07/2026

***Proposal/Active Project End Date: (MM/YYYY):** 03/2027

***Total Anticipated Proposal/Project Amount:** \$750,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	1.35
2027	0.68

***Overall Objectives:**

This Phase I Genesis project responds to Topic 18-D with a holistic approach to "Runtime Consciousness." Investigators at TN Tech and Argonne National Laboratory will deliver enhanced runtime optimization through AI-based methods that transform the potential for performance enhancement of DOE-relevant applications. The results of this work will be generalizabl across the DOE complex of large-scale applications.

***Statement of Potential Overlap:**

No overlap.

***Proposal/Active Project Title:** Collaborative Research: CyberTraining: Implementation: Small:Training the Next Generation of Data Facilitators for Secure and Trustworthy Modern Workflows

***Status of Support:** Pending

Proposal/Award Number:

***Source of Support:** NSF

***Primary Place of Performance:** Tennessee Technological University, Cookeville, TN

***Proposal/Active Project Start Date: (MM/YYYY):** 07/2026

***Proposal/Active Project End Date: (MM/YYYY):** 06/2029

***Total Anticipated Proposal/Project Amount:** \$285,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.25
2027	0.25
2028	0.25

***Overall Objectives:**

This proposal will develop practical curricular material and hands-on exercises that reflect day-to-day work of Data Facilitators focused on skill-gaps and including HPC operations, AI/ML workflow, workflow tooling, usable security, and facilitation skills. Two cohorts of four students each will trained from among universities across Tennessee (and receive stipends to support their in-service training). The investigators will work closely together to align training with nationally recognized frameworks and best practices. This cohort-based learning experience will encourage collaboration between and among the the participants, their institutions, and the investigators that extends beyond the period of performance.

***Statement of Potential Overlap:**

No overlap.

***Proposal/Active Project Title:** CS2: Highly Reliable MPI

***Status of Support:** Pending

Proposal/Award Number:

***Source of Support:** National Science Foundation

***Primary Place of Performance:** TN Tech, Cookeville, TN

***Proposal/Active Project Start Date: (MM/YYYY):** 07/2026

***Proposal/Active Project End Date: (MM/YYYY):** 06/2030

***Total Anticipated Proposal/Project Amount:** \$800,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.25
2027	0.25
2028	0.25
2029	0.25

***Overall Objectives:**

The purpose of this project is to create a collaborative research effort between TN TECH and LLNL scientists in pursuit of formal methods applied to resolving faults in parallel programs based on the Message Passing Interface (MPI). Asynchronous liquid separation (ALS) types are a key methodology applied in this research.

***Statement of Potential Overlap:**

No overlap.

***Proposal/Active Project Title:** OpenCCL: A Unified Open Standard for GPU Collective Communications

***Status of Support:** Pending

Proposal/Award Number:

***Source of Support:** DOE

***Primary Place of Performance:** Tennessee Tech, Cookeville, TN

***Proposal/Active Project Start Date: (MM/YYYY):** 07/2026

***Proposal/Active Project End Date: (MM/YYYY):** 03/2027

***Total Anticipated Proposal/Project Amount:** \$750,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	1
2027	0.4

***Overall Objectives:**

The key objective of this effort is to design, prototype, and evaluate OpenCCL, a vendor-neutral standard for GPU collective communications—analogue to MPI for CPU clusters—and by pairing that standard with an AI-driven porting engine that automatically transforms existing DOE codes to target it.

***Statement of Potential Overlap:**

No overlap.

***Proposal/Active Project Title:** Reducing the Time to Science while Enhancing Sustainability and Performance-Portability of DOE Flagship Applications at Exascale and Beyond

***Status of Support:** Pending

Proposal/Award Number:

***Source of Support:** DOE

***Primary Place of Performance:** Tennessee Tech, Cookeville, TN

***Proposal/Active Project Start Date: (MM/YYYY):** 07/2026

***Proposal/Active Project End Date: (MM/YYYY):** 03/2027

***Total Anticipated Proposal/Project Amount:** \$750,000

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.35
2027	0.15

***Overall Objectives:**

This Phase-I Genesis Project responds to Topic 18-A with transformative technology based on AI that enables the transformation of Fortran-based applications to C++ and modern parallelism constructs based on Kokkos with performance, productivity, and scalability goals. The results of this work will be generalizable across the DOE complex of large-scale applications.

***Statement of Potential Overlap:**

None.

***Proposal/Active Project Title:** Category III: FLAIR: Federated AI/ML-Ready Data Infrastructure for Intelligent and Scalable Data-Driven Workflows

***Status of Support:** Pending

Proposal/Award Number:

***Source of Support:** National Science Foundation

***Primary Place of Performance:** Tennessee Tech, Cookeville, TN

***Proposal/Active Project Start Date: (MM/YYYY):** 06/2026

***Proposal/Active Project End Date: (MM/YYYY):** 05/2028

***Total Anticipated Proposal/Project Amount:** \$498,780

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2026	0.1
2027	0.1

***Overall Objectives:**

The overall objective of this effort is to design a data fabric will provide a unified, intelligent data layer that lets researchers run large-scale, distributed AI/ML workflows across independent organizations without manually reconciling data formats, metadata, policies, or compute locations.

***Statement of Potential Overlap:**

No overlap.

***Proposal/Active Project Title:** MRI: Track-1: Acquisition of a Multi-accelerator Architecture for Advanced Scalable Computing Research

***Status of Support:** Pending

Proposal/Award Number:

***Source of Support:** National Science Foundation

***Primary Place of Performance:** Tennessee Technological University, Cookeville, TN

***Proposal/Active Project Start Date: (MM/YYYY):** 07/2025

***Proposal/Active Project End Date: (MM/YYYY):** 06/2028

***Total Anticipated Proposal/Project Amount:** \$1,333,501

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2025	0.01
2026	0.01

Year	Person Months
2027	0.01

***Overall Objectives:**

This MRI Acquisition Project is a multi-accelerator cluster system to explore the use of multiple types of heterogeneous processing elements in pursuit of several complementary science drivers. By working with a combined APU, FPGA, DPU system with dual networks, scientists and engineers collaborating on this system will be able to explore research in HPC systems, cybersecurity, and beyond.

***Statement of Potential Overlap:**

No overlap.

***Proposal/Active Project Title:** CC* Strategy-Region: Developing a Cyberinfrastructure framework for Collaborative Education and Research in Tennessee

***Status of Support:** Pending

Proposal/Award Number:

***Source of Support:** National Science Foundation

***Primary Place of Performance:** UTK, Knoxville TN and TNTECH, Cookeville, TN

***Proposal/Active Project Start Date: (MM/YYYY):** 03/2025

***Proposal/Active Project End Date: (MM/YYYY):** 02/2027

***Total Anticipated Proposal/Project Amount:** \$17,378

*** Person Months per budget period Devoted to the Proposal/Active Project:**

Year	Person Months
2025	0.18
2026	0.18

***Overall Objectives:**

To plan for regional computational and data infrastructure in support of the 80+ Institutions of Higher Learning in Tennessee.

***Statement of Potential Overlap:**

No overlap.

Addendum

Do the Notice of Funding Opportunity (NOFO) or Award instructions require a Current and Pending (Other) Support addendum?

No

Certification:

I understand that I have been designated as a covered individual by the Federal funding agency.

I certify to the best of my knowledge and belief that the information contained in this Current and Pending (Other) Support Form is true, complete, and accurate. I understand that any false, fictitious, or fraudulent information, misrepresentations, half-truths, or omissions of any material fact, may subject me to criminal, civil or administrative penalties for fraud, false statements, false claims or otherwise. (18 U.S.C. §§ 1001 and 287, and 31 U.S.C. 3729-3733 and 3801-3812). I further understand and agree that (1) the statements and representations made herein are material to DOE's funding decision, and (2) I have a responsibility to update the disclosures during the project period of the award should circumstances change which impact the responses provided above.

I also certify that, at the time of submission, I am not a party in a malign foreign talent recruitment program. I further understand should I take action to involve myself with a Malign Foreign Talent Recruitment Program during the period of performance of the award, I must notify the recipient's Authorized Agent immediately, but no later than five business days of taking such action and immediately recuse myself from all DOE awards.

If this is an R&D proposal/award, I also certify to the following statement:

Within the past 12 months I have completed research security training meeting the requirements in SEC. 10634(b) of 42 USC 19234.

Certified by Skjellum, Anthony in SciENCv on 2026-04-22 13:29:49